C++ FUNDAMENTALS

Essential Practice Problems & Exam Guide for Beginners

1. Basics (Input/Output & Arithmetic)

Problem 1.1: Student Grade & Fee Receipt

Write a program that takes a student's name, credit hours taken, and fee per credit hour. Calculate the total tuition fee and print a clean formatted summary receipt.

Hint: *Use std::cin and std::cout. Remember to declare variables with correct data types (string for name, double for fees).*

Sample I/O:

```
Input: John Doe, 12 credits, 150.00 MYR/credit
Output: Student: John Doe | Total Fee: 1800.00 MYR
```

EXAMPLE Code:

```
int main() {
    string studentName;
    int creditHours;
    double feePerCredit;

    // Taking input from the user
    cout << "Enter student name: ";
    getline(cin, studentName); // Allows names with spaces like "John Doe"

    cout << "Enter credit hours taken: ";
    cin >> creditHours;

    cout << "Enter fee per credit hour (MYR): ";
    cin >> feePerCredit;

    // Calculate total tuition fee
    double totalFee = creditHours * feePerCredit;

    cout << "\nOutput: Student: " << studentName << " | Total Fee: " << totalFee << " MYR" << endl;
    return 0; }
```

Problem 1.2: Ringgit to USD Converter

Write a program that takes an amount in Malaysian Ringgit (MYR) and converts it to US Dollars (USD) based on a fixed exchange rate (e.g., 1 USD = 4.45 MYR). Display the result rounded to 2 decimal places.

Hint: *Use fixed and setprecision(2) from <iomanip> library to format decimal output cleanly.*

Sample I/O:

```
Input: 445.00 MYR
Output: Equivalent in USD: $100.00
```

For helping you, code:

```
#include <iostream>
#include <iomanip> // Required for fixed and setprecision
using namespace std;

int main() {
    double myrAmount, usdAmount;
    const double EXCHANGE_RATE = 4.45; // 1 USD = 4.45 MYR

    // Take input from the user
    cout << "Input: ";
    cin >> myrAmount;

    // Perform conversion
    usdAmount = myrAmount / EXCHANGE_RATE;

    // Display the result formatted to 2 decimal places
    cout << fixed << setprecision(2);
    cout << "Output: Equivalent in USD: $" << usdAmount << endl;

    return 0;
}
```

Note: fixed and setprecision(2) to force the decimal numbers to always display cleanly with exactly 2 digits after the decimal point.

2. Conditionals (if-else)

Problem 2.1: University Grading System

Write a program that takes a student's final exam score (0 to 100) and assigns a grade: 'A' (80-100), 'B' (65-79), 'C' (50-64), or 'F' (Below 50). Include validation to check if the score is within valid bounds.

Hint: *Use chained/Nested if-else if statements. Check for invalid scores (<0 or >100) first before grading.*

Sample I/O:

```
Input: 72
Output: Grade: B
```

Problem 2.2: Movie Ticket Discount Eligibility

Write a program that determines movie ticket pricing based on age and student status. Children under 12 and seniors 60+ get a discount. University students with a valid ID also get a student discount.

Hint: *Combine logical operators (&&, ||) carefully in your condition checks.*

Sample I/O:

```
Input: Age = 21, Student ID = Yes
Output: Eligible for Student Discount. Price: 12.00 MYR
```

3. Switch Case

Problem 3.1: Campus Cafeteria Menu Selector

Create a program with a menu: 1. Nasi Lemak, 2. Teh Tarik, 3. Roti Canai, 4. Burger. Ask the user to enter a choice number and display the selected item along with its standard price.

Hint: *Don't forget the 'break' statement after each case to prevent fall-through bugs.*

Sample I/O:

```
Input: Choice = 2
Output: You selected Teh Tarik. Price: 2.50 MYR
```

Problem 3.2: Simple Arithmetic Calculator

Write a program that takes two numbers and an operator character (+, -, *, /) using a switch statement to perform the corresponding operation and display the result.

Hint: *Handle division by zero explicitly in the division case ('/') to prevent runtime crashes.*

Sample I/O:

```
Input: 10 + 5
Output: Result = 15
```

4. Loops (for, while)

Problem 4.1: Monthly Savings Goal Tracker

Write a program that asks a student how much they want to save. Using a loop, prompt them to enter their weekly savings for 4 weeks, calculate the total saved, and check if they reached their goal.

Hint: *Use a 'for' loop running 4 times with an accumulator variable (e.g., total += weekly_amount).*

Sample I/O:

```
Input: Goal = 200 | Weekly: 50, 40, 60, 50
Output: Total Saved: 200 MYR. Goal Achieved!
```

Problem 4.2: Multiplication Table Generator

Write a program that takes an integer N from the user and prints its multiplication table from 1 up to 10 in standard tabular format.

Hint: *A simple 'for' loop from i = 1 to 10 multiplying N by i will do the trick.*

Sample I/O:

```
Input: 5
Output: 5 x 1 = 5 ... 5 x 10 = 50
```

5. Arrays (1D)

Problem 5.1: Semester CGPA Subject Marks Tracker

Write a program that takes marks for 5 subjects into an array, computes the average mark, and finds the highest mark obtained.

Hint: *Loop through the array once to calculate the sum/average, and use a comparison variable to track the maximum value.*

Sample I/O:

```
Input: Marks: 75, 82, 90, 65, 88
Output: Average: 80.0 | Highest Mark: 90
```

Problem 5.2: Passing vs Failing Student Count

Given an array of exam scores for 6 students, write a program to count how many students passed (score >= 50) and how many failed (score < 50).

Hint: *Initialize two counter variables to zero and increment them inside a loop checking each array element.*

Sample I/O:

```
Input: Scores: 45, 60, 80, 35, 50, 92
Output: Passed: 4 | Failed: 2
```

6. Functions

Problem 6.1: Function for Rectangle Area & Perimeter

Write two separate functions: calculateArea(length, width) and calculatePerimeter(length, width). In main(), ask the user for dimensions and call both functions.

Hint: *Make sure your functions return a value (e.g., double) and accept appropriate parameters.*

Sample I/O:

```
Input: Length = 5.0, Width = 3.0
Output: Area = 15.0 | Perimeter = 16.0
```

Problem 6.2: Discount Price Calculator Function

Write a function named calculateDiscountedPrice that takes the original price of an item (as a double) and a discount percentage (as a double), and returns the final price after applying the discount. Demonstrate its usage in main() by asking the user for input and printing the result.

Hint: Calculate the discount amount by multiplying originalPrice * (discountPercent / 100.0), then subtract it from the original price. Make sure your function returns a double.

Sample I/O:

```
Input: Original Price = 150.00 MYR, Discount = 20%
Output: Final Price after discount: 120.00 MYR
```